

Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Computer Science researcher and research engineer focused on **trustworthy agentic AI**, **long-horizon multi-agent systems**, **empirical evaluation**, and **reliable AI workflows**.

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Canada

Completed Summer 2024

Selected Research and Engineering Projects

Codex Boss — *Multi-agent orchestration and evidence-based acceptance platform*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Building a platform for long-running AI workflows in which multiple agents can plan, execute, critique, and adjudicate work while preserving durable task state.
- Developed platform foundations spanning durable state/events, capability boundaries, plugin isolation, knowledge/data lifecycle, dogfooding, and acceptance/verification.
- Investigating how autonomous systems can distinguish meaningful task completion from superficially plausible but weak evidence.
- Treating unresolved failure cases as evidence for further evaluation rather than reporting an unfinished acceptance mechanism as solved.

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building an execution layer for unattended software work with optional capabilities isolated as extensions.
- Implemented ordered/scheduled task queues, hardware-adaptive local concurrency, unified task lifecycle and terminal events, task history, and terminal notifications.
- Designed extension-failure isolation, restart/recovery-oriented execution, and an idempotent Windows installer with explicit optional-plugin flow.
- Using the system as a testbed for persistent execution, failure recovery, scheduling, and completion verification in long-running software-agent workflows.

Research Interests

Trustworthy agentic AI; long-horizon agent orchestration; multi-agent reliability; failure propagation and recovery; empirical agent evaluation; reproducible benchmarks for tool-using AI.

Technical and Research Strengths

- End-to-end research engineering: problem formulation, system design, iterative implementation, testing, debugging, and empirical evaluation.
- Multi-module software design with explicit persistence, lifecycle, extensibility, isolation, recovery, and acceptance criteria.
- Git/GitHub-based experimental development, branch-based iteration, regression testing, and reproducible project documentation.
- Practical multi-AI workflow design while independently reviewing plans, evidence, failure modes, and completion claims.

Current Research Direction

I am interested in **trustworthy agentic AI systems**: how tool-using agents should preserve state, coordinate actions and evidence, recover from partial failures, and be evaluated over long horizons where small errors can compound. I would like to study these questions through system building, realistic task benchmarks, and empirical failure analysis.